

In lecture,  $SE_2(3) = SO(3) \ltimes (\mathbb{R}^3 \times \mathbb{R}^3)$  was introduced for 3D inertial navigation, combining orientation, velocity, and position into a single Lie group. You will now derive the **2D analog**,  $SE_2(2) = SO(2) \ltimes (\mathbb{R}^2 \times \mathbb{R}^2)$ , which adds velocity to the  $SE(2)$  pose.